Solution 1: Entropy in Binary Classification

(a) By definition, the entropy of a Bernoulli random variable is:

$$\begin{aligned} H(\epsilon) &= -\sum_{e \in \{0,1\}} \mathbb{P}(\epsilon = e) \log_2(\mathbb{P}(\epsilon = e)) \\ &= -\theta_\epsilon \log_2(\theta_\epsilon) - (1 - \theta_\epsilon) \log_2(1 - \theta_\epsilon) \end{aligned}$$

(b) The conditional entropy quantifies the uncertainty of Y if the outcome of X is given. It is defined as the expected value of the entropies of the conditional distributions, averaged over the conditioning random variable.

$$\begin{aligned} H(Y|X) &= \mathbb{E}_X[H(Y|X = x)] = \sum_{x \in \mathcal{X}} \mathbb{P}(X = x) H(Y|X = x) \\ &= \theta_X H(Y|X = 1) + (1 - \theta_X) H(Y|X = 0) \end{aligned}$$

Let's think about what happens with Y when the values of X are given.

- If $X = 1$, the maximum between ϵ and X will always be 1 and Y will always be 1. As there is no uncertainty, the conditional entropy is 0.
- If $X = 0$, then $Y = 2\epsilon - 1$, which can be -1 or 1. Since X and ϵ are independent, ϵ given $X = 0$ has the same distribution as ϵ. As $\epsilon \mapsto 2\epsilon - 1$ is bijective, it does not change the entropy, so $H(Y|X = 0) = H(\epsilon)$.

$$\begin{aligned} H(Y|X) &= \theta_X \underbrace{H(Y|X = 1)}_{=0} + (1 - \theta_X) \underbrace{H(Y|X = 0)}_{=H(\epsilon)} \\ &= (1 - \theta_X) H(\epsilon) \end{aligned}$$

(c) Using the chain rule for entropy:

$$\begin{aligned} H(Y, X) &= H(X) + H(Y|X) \\ &= -\theta_X \log_2(\theta_X) - (1 - \theta_X)(\log_2(1 - \theta_X) - H(\epsilon)) \end{aligned}$$

(d) Now ϵ has a deterministic relation with X, so X and ϵ are no longer independent (unless ψ is constant). Let's see how our results change with this modification.

- As $\epsilon = \psi(X)$ is a deterministic function of X, ψ introduces no additional uncertainty. Hence, the entropy of ϵ cannot exceed the entropy of X:

  $$H(\epsilon) = H(\psi(X)) \leq H(X)$$

  Since $\psi : \{0, 1\} \to \{0, 1\}$, there are only two cases: if ψ is bijective ($\psi(x) = x$ or $\psi(x) = 1 - x$), then $H(\epsilon) = H(X)$; if ψ is constant, then ϵ is not random and $H(\epsilon) = 0$.

- Y is a function of X and ϵ, and now ϵ has a deterministic relation with X. Accordingly, we can conclude that Y is now a function of X.

$$Y = 2\max\{X, \psi(X)\} - 1$$

If the value of X is given, there is no uncertainty associated.

$$H(Y|X) = 0$$

- Using the chain rule for entropy:

$$\begin{aligned} H(Y, X) &= H(X) + \underbrace{H(Y|X)}_{=0} \\ &= H(X) \end{aligned}$$

This result is expected. As X and Y have a deterministic relation, the uncertainty is only given by X.